



US 20250279291A1

(19) **United States**

(12) **Patent Application Publication**
LIEN et al.

(10) **Pub. No.: US 2025/0279291 A1**

(43) **Pub. Date: Sep. 4, 2025**

(54) **EXHAUST PIPING FOR A
SEMICONDUCTOR PROCESSING TOOL**

(71) Applicant: **Taiwan Semiconductor
Manufacturing Company, Ltd.,
Hsinchu (TW)**

(72) Inventors: **Kuo-Sheng LIEN, Hsinchu (TW);
Yung-Ho CHEN, Hsinchu City (TW)**

(21) Appl. No.: **19/211,605**

(22) Filed: **May 19, 2025**

Related U.S. Application Data

(63) Continuation of application No. 17/446,580, filed on
Aug. 31, 2021, now Pat. No. 12,334,365.

Publication Classification

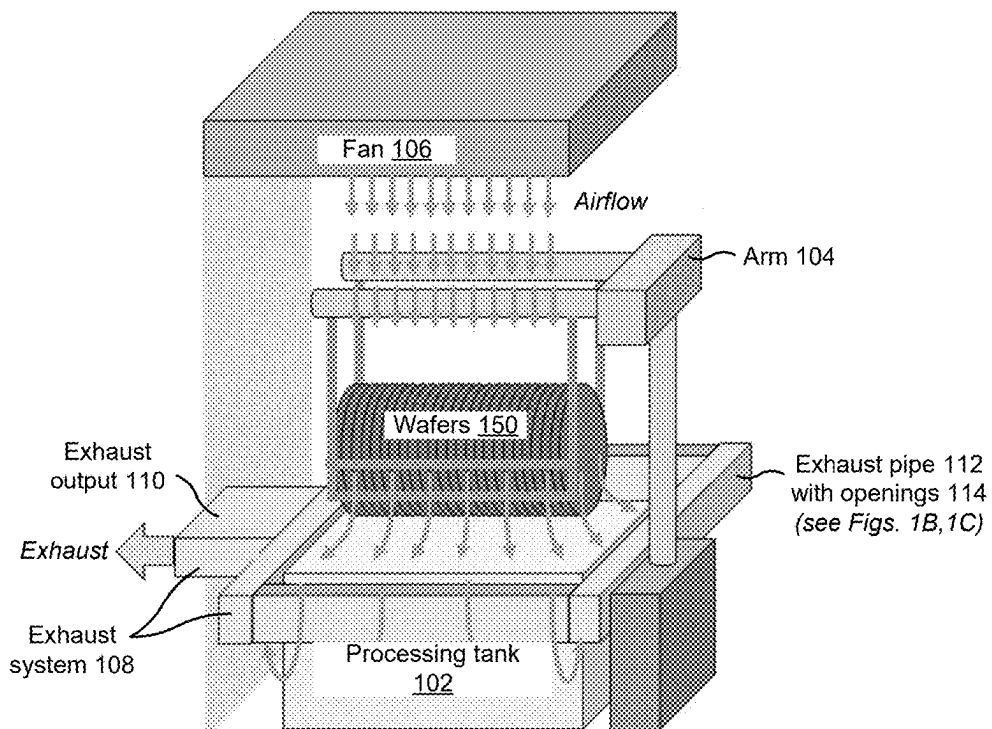
(51) **Int. Cl.**
H01L 21/67 (2006.01)
B08B 3/08 (2006.01)
B08B 13/00 (2006.01)
F26B 3/04 (2006.01)
F26B 9/10 (2006.01)
F26B 21/00 (2006.01)
H01L 21/677 (2006.01)

(52) **U.S. Cl.**
CPC **H01L 21/67023** (2013.01); **B08B 3/08**
(2013.01); **B08B 13/00** (2013.01); **F26B 3/04**
(2013.01); **F26B 9/10** (2013.01); **F26B 21/004**
(2013.01); **H01L 21/67017** (2013.01); **H01L**
21/67028 (2013.01); **H01L 21/6704** (2013.01);
H01L 21/67057 (2013.01); **H01L 21/67075**
(2013.01); **H01L 21/67086** (2013.01); **H01L**
21/67757 (2013.01)

(57) **ABSTRACT**

A semiconductor processing tool may include a processing tank. The semiconductor processing tool may include an arm arranged to hold a plurality of wafers over the processing tank such that wafers in the plurality of wafers are horizontally stacked over the processing tank. The semiconductor processing tool may include a fan arranged over the arm to permit an airflow to be provided across surfaces of the wafers in a vertical direction toward the processing tank. The semiconductor processing tool may include an exhaust system including at least one exhaust output and one or more exhaust pipe segments arranged substantially around the processing tank and connected to the at least one exhaust output. The one or more exhaust pipe segments may include a plurality of openings to receive exhaust air to be provided to the at least one exhaust output.

100 →



100 →

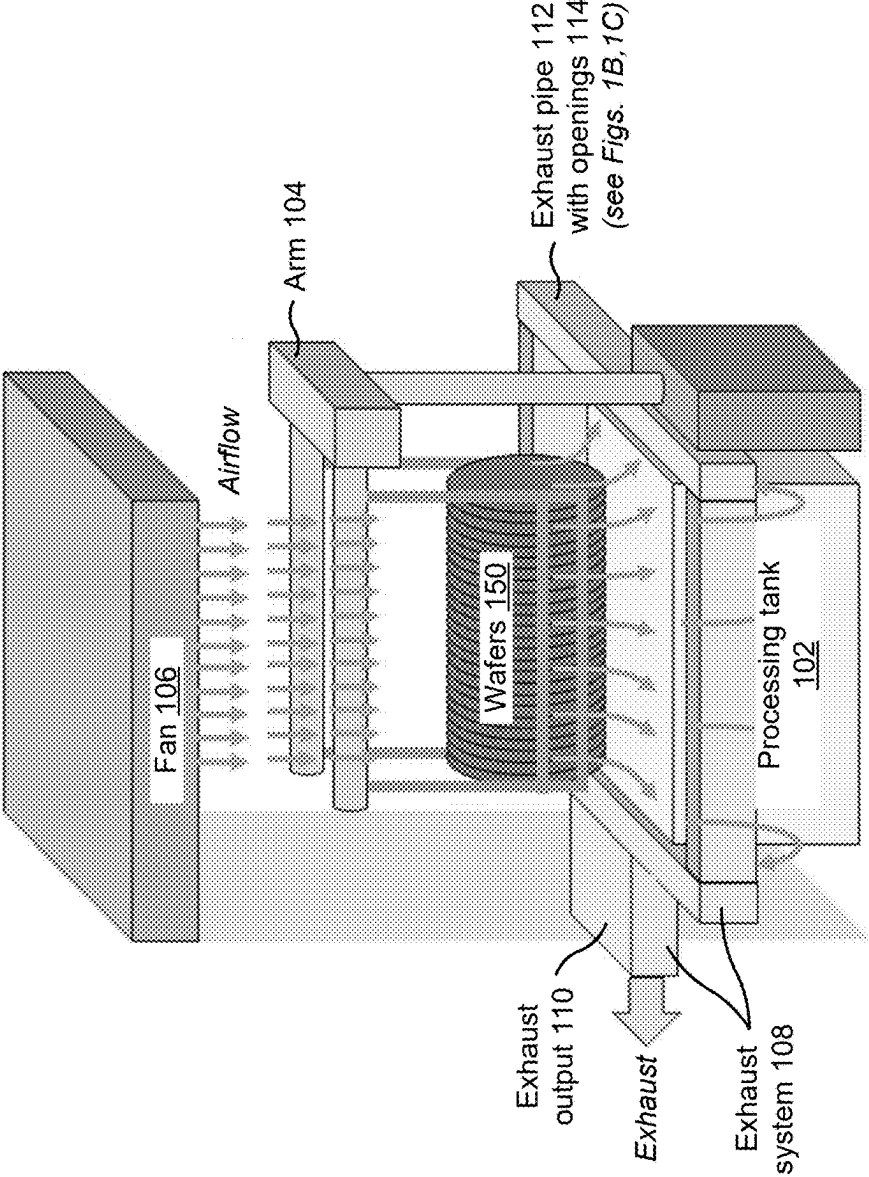


FIG. 1A

100 →

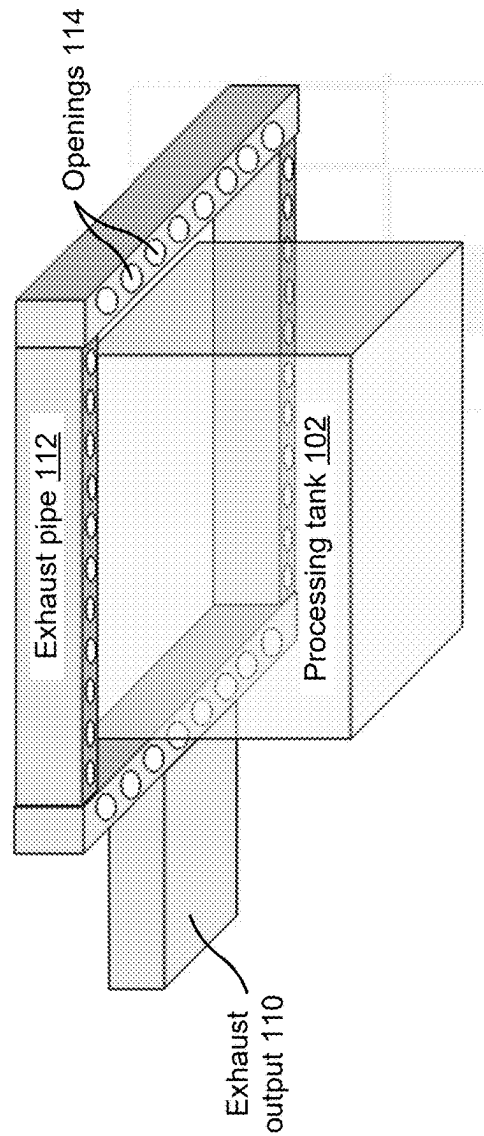


FIG. 1B

100 →

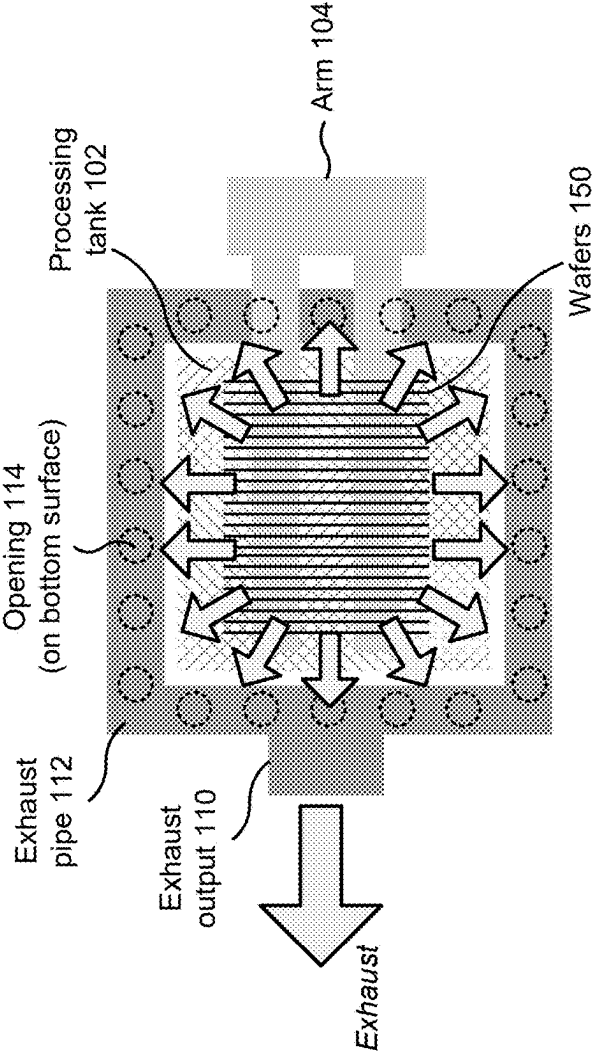


FIG. 1C

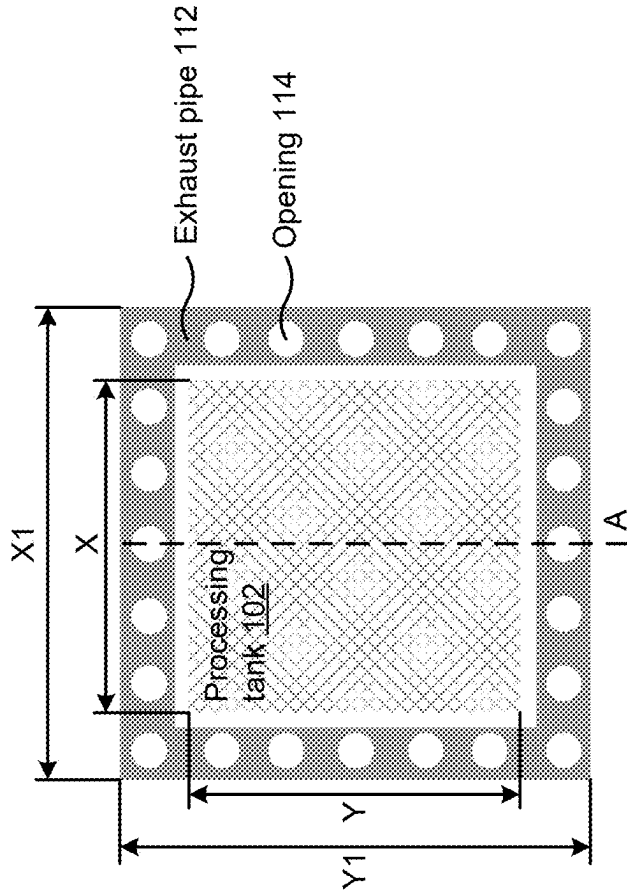


FIG. 2A

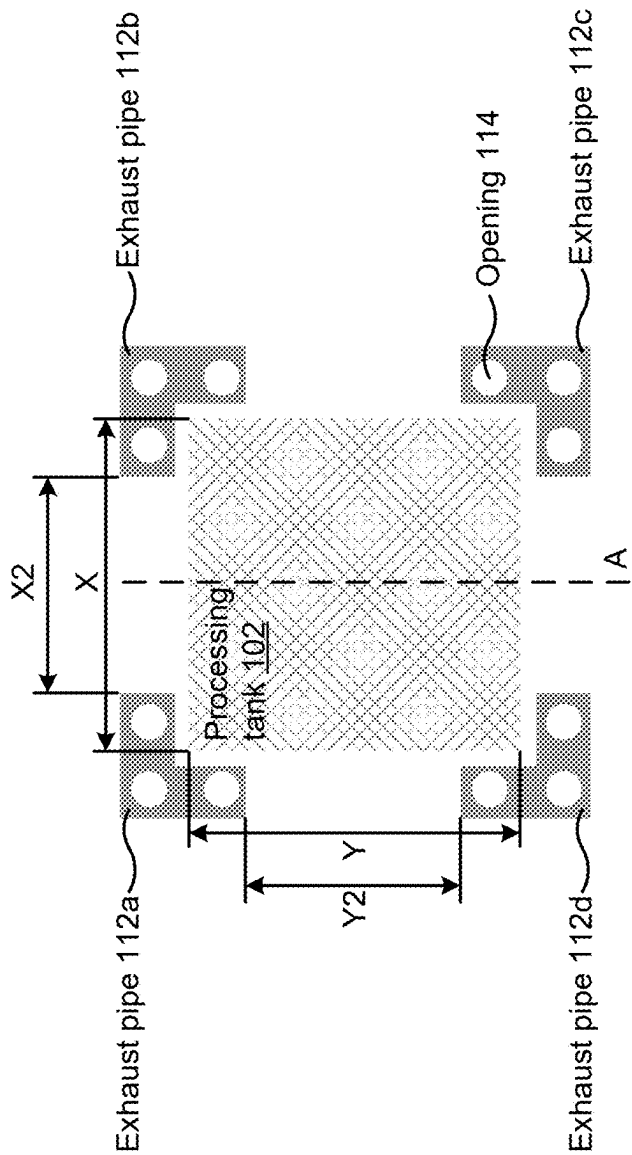


FIG. 2B

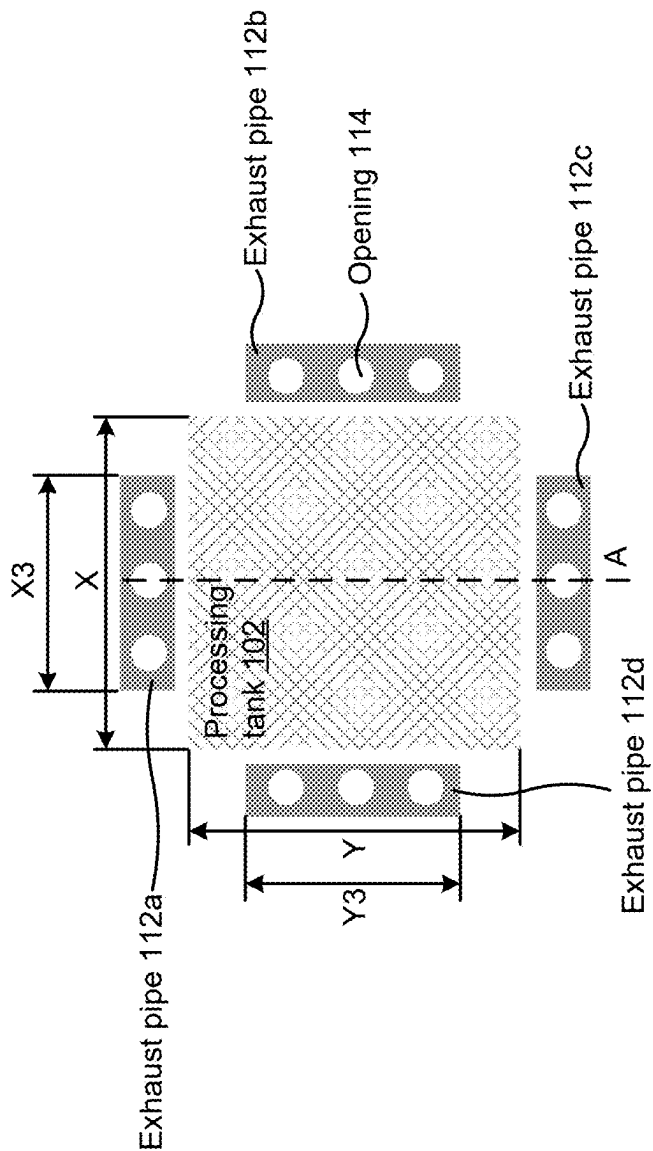


FIG. 2C

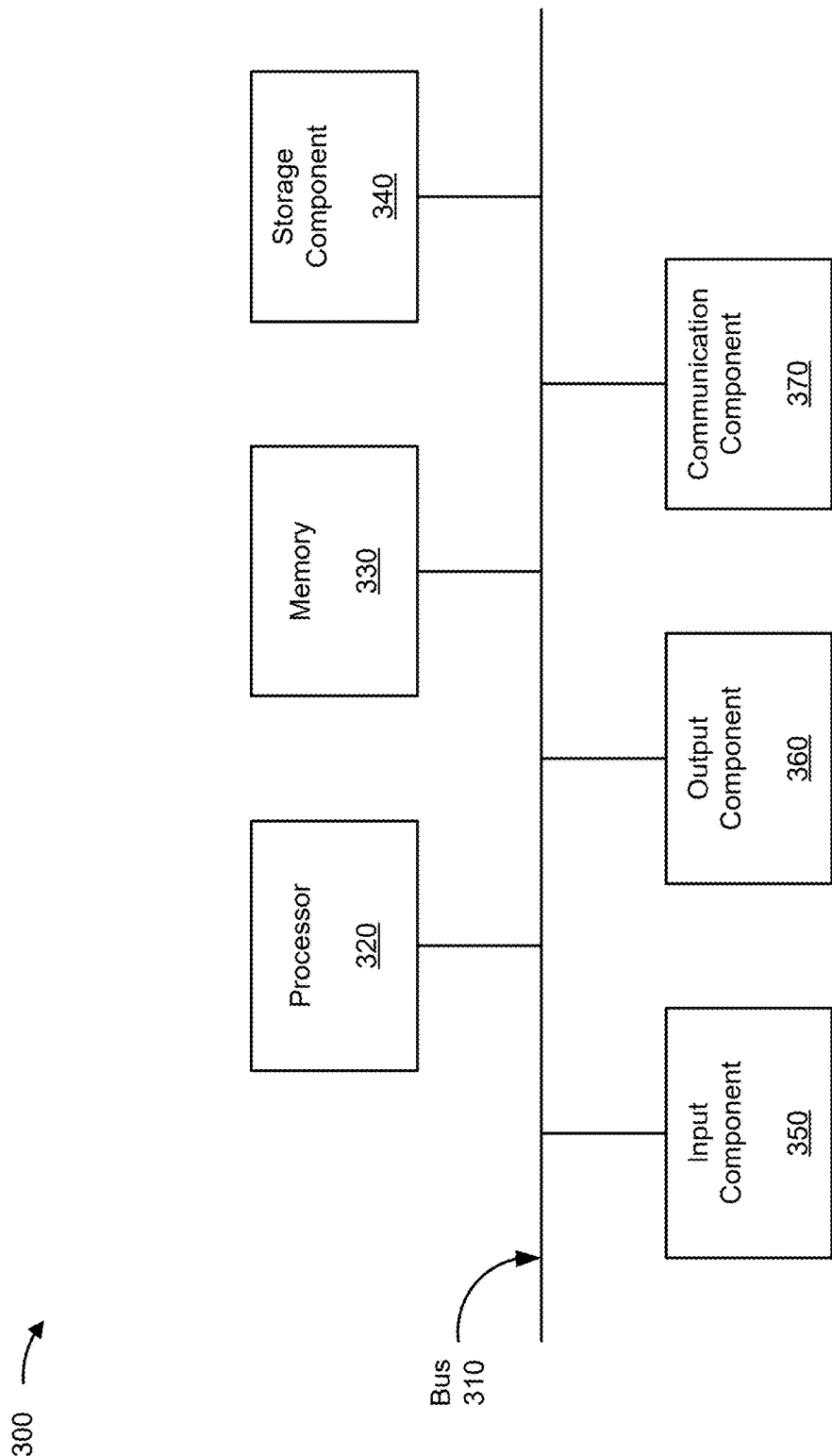


FIG. 3

EXHAUST PIPING FOR A SEMICONDUCTOR PROCESSING TOOL

RELATED APPLICATION

[0001] This application is a continuation of U.S. patent application Ser. No. 17/446,580, filed Aug. 31, 2021, which is incorporated herein by reference in its entirety.

BACKGROUND

[0002] A semiconductor processing tool, such as a wet bench, may include a fan used for drying a group of wafers after processing by the semiconductor processing tool (e.g., after wet cleaning, after wet etching, or the like). The fan may, for example, be mounted over a position at which a stack of wafers is horizontally held by an arm of the semiconductor processing tool. Here, the arrangement of the fan permits the fan to blow air across the top and bottom surfaces of the wafers in the stack.

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0004] FIGS. 1A-1C are diagrams of an example implementation of a semiconductor processing tool including an exhaust system, as described herein.

[0005] FIGS. 2A-2C are diagrams illustrating various example arrangements of exhaust pipe segments of the exhaust system described in connection with FIGS. 1A-1C.

[0006] FIG. 3 is a diagram of example components of one or more devices of a semiconductor processing tool including an exhaust system, as described herein.

DETAILED DESCRIPTION

[0007] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0008] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90

degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0009] In some cases, a semiconductor processing tool including a fan used for drying wafers (e.g., after processing) also includes an exhaust output for drawing exhaust air from the semiconductor processing tool. The exhaust output is mounted on one side of a processing tank of the semiconductor processing tool to draw the exhaust air as air is blown over the wafers by the fan. However, the exhaust output being mounted on only one side of the semiconductor processing tool causes air to flow generally in a single direction (toward an intake of the exhaust output). This unidirectional air flow can result in a concentration of acid gas and/or a moisture concentration being uneven among wafers after drying, which can increase a defect rate in wafers processed by the semiconductor processing tool and lead to increased costs and additional operations by this and/or one or more other semiconductor processing tools to address defective wafers.

[0010] Some implementations described herein provide an exhaust system for a semiconductor processing tool. In some implementations, the exhaust system includes at least one exhaust output, and one or more exhaust pipe segments arranged substantially around the processing tank. The one or more exhaust pipe segments include openings to receive exhaust air to be provided to the at least one exhaust output. In some implementations, the exhaust system being substantially around the processing tank (e.g., 360 degrees around) causes airflow to be uniformly distributed within the environment of the semiconductor processing tool (e.g., such that the air flows in all directions around the wafers, rather than a single direction). As a result, a concentration of acid gas and/or a moisture concentration may be relatively even among wafers in the stack, thereby decreasing a defect rate in wafers processed by the semiconductor processing tool and reduce costs and additional operations that would have otherwise been performed by this and/or one or more other semiconductor processing tools to address defective wafers.

[0011] FIGS. 1A-1C are diagrams of an example implementation of a semiconductor processing tool **100** including an exhaust system **108**, as described herein. The semiconductor processing tool **100** may be, for example, a wet bench designed to perform wet cleaning, wet etching, or another type of semiconductor processing operation.

[0012] As shown in FIGS. 1A-1C, the semiconductor processing tool **100** includes a processing tank **102**, an arm **104**, a fan **106**, and the exhaust system **108**, where the exhaust system **108** includes at least one exhaust output **110** and one or more exhaust pipe segments **112**. As described in further detail below, the one or more exhaust pipe segments **112** include a plurality of openings **114**. FIG. 1A illustrates an isometric view of the semiconductor processing tool **100** from above the exhaust system **108**, FIG. 1B illustrates an isometric view of the processing tank **102** and the exhaust system **108** from below the exhaust system **108**, and FIG. 1C is a two-dimensional (2D) top view of the semiconductor processing tool **100** from above the arm **104**.

[0013] The processing tank **102** is a component associated with performing a semiconductor processing operation on one or more wafers **150** (e.g., one or more semiconductor wafers). For example, the processing tank **102** may be a tank that stores a solution (e.g., a liquid solution) associated with wet cleaning the one or more wafers **150**, wet etching the

one or more wafers 150, or the like. In some implementations, the one or more wafers 150 may be held above the processing tank 102 while the fan 106 provides an airflow toward the one or more wafers 150 (e.g., in association with drying the one or more wafers 150).

[0014] The arm 104 is a component to move, hold, or otherwise control a position of one or more wafers 150 in association with the semiconductor process operation performed by the semiconductor processing tool 100. For example, the arm 104 may be capable of grabbing a plurality of wafers 150 (e.g., a stack of wafers in a wafer carrier), submerging the plurality of wafers 150 a solution in the processing tank 102, removing the plurality of wafers 150 from the solution, and holding the plurality of wafers 150 over the processing tank 102 (e.g., during drying of the plurality of wafers 150). In some implementations, as indicated in FIG. 1A, the arm 104 may be arranged to permit the arm 104 to hold the plurality of wafers 150 such that wafers 150 in the plurality of wafers 150 are horizontally stacked over the processing tank 102 (e.g., such that top and bottom surfaces of each of the plurality of wafers 150 are oriented in a vertical direction).

[0015] The fan 106 is a component to provide an airflow across surfaces of the plurality of wafers 150 (e.g., in association with drying the wafers 150). For example, the fan 106 may provide an airflow across (e.g., parallel to) top and bottom surfaces of wafers 150 of the plurality of wafers 150 while the arm 104 holds the plurality of wafers 150 over the processing tank 102 after the arm 104 has removed the plurality of wafers 150 from the processing tank 102. In some implementations, as shown in FIG. 1A, the fan 106 is arranged over the arm 104 to permit the airflow to be provided across the surfaces of the plurality of wafers 150 in a vertical direction (e.g., a downward direction from the fan 106 toward the processing tank 102).

[0016] The exhaust system 108 is a component to draw exhaust air from the environment of the semiconductor processing tool 100 and direct the exhaust air away from the semiconductor processing tool 100. In some implementations, the exhaust system 108 may draw the exhaust air away from the environment of the semiconductor processing tool 100 concurrently with the fan 106 providing the airflow over the surfaces of the plurality of wafers 150 (e.g., such that air flows over the surfaces of the plurality of wafers 150 and is drawn away by the exhaust system 108). As shown, the exhaust system 108 may include at least one exhaust output 110 and one or more exhaust pipe segments 112.

[0017] The exhaust output 110 is a component to receive exhaust air from one or more exhaust pipe segments 112 and direct the exhaust air away from the environment of the semiconductor processing tool 100. In some implementations, as illustrated in FIGS. 1A-1C, the exhaust system 108 may include a single exhaust output 110, and each of one or more exhaust pipe segments 112 of the exhaust system 108 are connected to the single exhaust output 110. Alternatively, in some implementations, the exhaust system 108 may include a plurality of exhaust outputs, and at least one exhaust pipe segment 112 of the one or more exhaust pipe segments 112 is connected to each exhaust output 110 of the plurality of exhaust outputs 110. For example, when the one or more exhaust pipe segments 112 include a first exhaust pipe segment 112 and a second exhaust pipe segment 112 that are not connected to one another, the first exhaust pipe segment 112 may be connected to a first exhaust output 110

on a first side of the processing tank 102 and the second exhaust pipe segment 112 may be connected to a second exhaust output 110 on a second (e.g., different, such as opposite) side of the processing tank 102.

[0018] The exhaust pipe segment 112 is a component to receive the exhaust air from the environment of the semiconductor processing tool 100 and provide the exhaust air to an exhaust output 110 (e.g., such that the exhaust air can be directed away from the environment of the semiconductor processing tool 100 by the exhaust output 110).

[0019] In some implementations, the one or more exhaust pipe segments 112 of the exhaust system 108 are arranged substantially around (e.g., on at least two sides of) the processing tank 102. For example, the one or more exhaust pipe segments 112 may be arranged 360 degrees around the processing tank 102. In the particular example shown in FIGS. 1A-1C, the one or more exhaust pipe segments 112 are arranged around an entirety of the perimeter of the processing tank 102. Additional details regarding the arrangement of the one or more exhaust pipe segments 112 are described below with respect to FIGS. 2A-2C.

[0020] In some implementations, the one or more exhaust pipe segments 112 include a plurality of openings 114 to receive exhaust air to be provided to the at least one exhaust output 110. In some implementations, the plurality of openings 114 may be arranged substantially around the perimeter of the processing tank 102. For example, the plurality of openings 114 may be (evenly) distributed around the perimeter of the processing tank 102, as shown in FIGS. 1B and 1C. In some implementations, the plurality of openings 114 is symmetrically arranged with respect to a center axis on a plane corresponding to the processing tank 102. In some implementations, such an arrangement of the openings 114 causes airflow within the environment of the semiconductor processing tool to be relatively uniformly distributed because exhaust air is drawn into the openings 114 at multiple locations substantially around the perimeter of the processing tank 102 (e.g., rather than at single location as with prior exhaust outputs). As a result, air flows in all directions around wafers 150 being dried in the semiconductor processing tool 100, which results in a more even concentration of acid gas and/or a moisture concentration among the wafers 150, thereby decreasing a defect rate in wafers processed by the semiconductor processing tool 100.

[0021] In some implementations, one or more openings 114 may be on a bottom surface of an exhaust pipe segment 112. That is, in some implementations, one or more openings 114 may be on a surface of the exhaust pipe segment 112 that faces a bottom of the processing tank 102, or on a surface of the exhaust pipe segment 112 that faces away from the fan 106. In some implementations, an opening 114 being on the bottom surface of the exhaust pipe segment 112 may act to prevent material (e.g., liquid droplets) from entering the opening 114 that could damage the exhaust pipe segment 112. In some implementations, one or more openings 114 may be on a side surface of an exhaust pipe segment 112 of the one or more exhaust pipe segments 112 (e.g., a surface facing a side of the processing tank 102, a surface opposite to the surface facing the side of the processing tank 102, or the like). In some implementations, one or more openings 114 may be on a top surface of an exhaust pipe segment 112 of the one or more exhaust pipe segments 112 (e.g., a surface facing toward the fan 106). In some implementations, an exhaust pipe segment 112 may include one or more openings

114 on a bottom surface of the exhaust pipe segment **112**, one or more openings **114** on a side surface of the exhaust pipe segment **112**, and/or one or more openings **114** on a top surface of the exhaust pipe segment **112**.

[0022] As indicated above, FIGS. 1A-1C are provided as an example. Other examples may differ from what is described with regard to FIGS. 1A-1C. The number and arrangement of components shown in FIGS. 1A-1C are provided as an example. In practice, there may be additional components, fewer components, different components, or differently arranged components than those shown in FIGS. 1A-1C. Furthermore, two or more components shown in FIGS. 1A-1C may be implemented within a single component, or a single component shown in FIGS. 1A-1C may be implemented as multiple, distributed components. Additionally, or alternatively, a set of components (e.g., one or more components) shown in FIGS. 1A-1C may perform one or more functions described as being performed by another set of components shown in FIGS. 1A-1C.

[0023] FIGS. 2A-2C are diagrams illustrating various example arrangements of exhaust pipe segments **112** of the exhaust system **108**. FIGS. 2A-2C illustrate 2D bottom views of the exhaust pipe segments **112** of the exhaust system **108**. For purposes of clarity, only the one or more exhaust pipe segments **112** (with openings **114**) and the processing tank **102** are shown in FIGS. 2A-2C.

[0024] In some implementations, the one or more exhaust pipe segments **112** consist of a single exhaust pipe segment **112** that is substantially around (e.g., on at least two sides of) a perimeter of the processing tank **102**. FIG. 2A is a diagram illustrating an example of a single exhaust pipe segment **112** that is around an entirety of the perimeter of the processing tank **102**. Notably, in the example shown in FIG. 2A, a pattern of the plurality of openings **114** is substantially symmetrical with respect to a center axis A on a plane corresponding to the processing tank **102**. In some implementations, as illustrated in FIG. 2A, a length of a given side of the exhaust pipe segment **112** may be greater than or equal to a length of a corresponding side of the processing tank **102** (e.g., X_1 may be greater than or equal to X and/or Y_1 may be greater than or equal to Y).

[0025] In some implementations, the one or more exhaust pipe segments **112** includes a plurality of exhaust pipe segments **112** arranged such that a gap is present along a side of the processing tank **102** between two of the plurality of exhaust pipe segments **112**. For example, as shown in FIG. 2B, exhaust pipe segments **112a** through **112d** may be arranged around the processing tank **102** such that a first gap is between exhaust pipe segments **112a** and **112b** along a first side of the processing tank **102**, a second gap is between exhaust pipe segments **112b** and **112c** along a second side of the processing tank **102**, a third gap is between exhaust pipe segments **112c** and **112d** along a third side of the processing tank **102**, and a fourth gap is between exhaust pipe segments **112d** and **112a** along a fourth side of the processing tank **102**. In some implementations, as illustrated in FIG. 2B, a length of a gap along a side of the processing tank **102** may be less than or equal to a length of a corresponding side of the processing tank **102** (e.g., X_2 may be less than or equal to X and/or Y_2 may be less than or equal to Y). Notably, in the example shown in FIG. 2B, a pattern of the plurality of openings **114** is substantially symmetrical with respect to a center axis A on a plane corresponding to the processing tank **102**. In some implementations, the exhaust pipe segments

112a through **112d** may be connected to a single exhaust output **110** or to different exhaust outputs **110**.

[0026] In some implementations, the one or more exhaust pipe segments **112** includes a plurality of exhaust pipe segments **112** arranged such that a gap is present at a corner of the processing tank **102** between two of the plurality of exhaust pipe segments **112**. For example, as shown in FIG. 2C, exhaust pipe segments **112a** through **112d** may be arranged around the processing tank **102** such that a first gap is between exhaust pipe segments **112a** and **112b** at a first corner of the processing tank **102**, a second gap is between exhaust pipe segments **112b** and **112c** at a second corner of the processing tank **102**, a third gap is between exhaust pipe segments **112c** and **112d** at a third corner of the processing tank **102**, and a fourth gap is between exhaust pipe segments **112d** and **112a** at a fourth corner of the processing tank **102**. In some implementations, as illustrated in FIG. 2C, a length of an exhaust pipe segment **112** along a side of the processing tank **102** may be less than or equal to a length of a corresponding side of the processing tank **102** (e.g., X_3 may be less than or equal to X and/or Y_3 may be less than or equal to Y). Notably, in the example shown in FIG. 2C, a pattern of the plurality of openings **114** is substantially symmetrical with respect to a center axis A on a plane corresponding to the processing tank **102**. In some implementations, the exhaust pipe segments **112a** through **112d** may be connected to a single exhaust output **110** or to different exhaust outputs **110**.

[0027] Notably, a combination of feature shown in FIGS. 2A-2C may be present in a given exhaust system **108**. For example, in some implementations, a length of an exhaust pipe segment **112** on a first side of the processing tank **102** may be greater than or equal to a length of the first side of the processing tank **102** (e.g., $X_1 > X$), while a length of a gap along a second side of the processing tank **102** may be less than or equal to a length of a second side of the processing tank **102** (e.g., $Y_2 \leq Y$). As another example, in some implementations, a length of an exhaust pipe segment **112** on a first side of the processing tank **102** may be greater than or equal to a length of the first side of the processing tank **102** (e.g., $X_1 > X$), while a length of an exhaust pipe segment **112** along a second side of the processing tank **102** may be less than or equal to a length of a second side of the processing tank **102** (e.g., $Y_3 \leq Y$). As another example, in some implementations, a length of a gap along a first side of the processing tank **102** may be less than or equal to a length of a first side of the processing tank **102** (e.g., $X_2 \leq X$), while a length of an exhaust pipe segment **112** along a second side of the processing tank **102** may be less than or equal to a length of a second side of the processing tank **102** (e.g., $Y_3 \leq Y$).

[0028] As indicated above, FIGS. 2A-2C are provided as an example. Other examples may differ from what is described with regard to FIGS. 2A-2C. The number and arrangement of components shown in FIGS. 2A-2C are provided as an example. In practice, there may be additional components, fewer components, different components, or differently arranged components than those shown in FIGS. 2A-2C. Furthermore, two or more components shown in FIGS. 2A-2C may be implemented within a single component, or a single component shown in FIGS. 2A-2C may be implemented as multiple, distributed components. Additionally, or alternatively, a set of components (e.g., one or more components) shown in FIGS. 2A-2C may perform one or

more functions described as being performed by another set of components shown in FIGS. 2A-2C

[0029] FIG. 3 is a diagram of example components of a device 300, which may correspond to the semiconductor processing tool 100 and/or one or more components of the semiconductor processing tool 100 (e.g., processing tank 102, arm 104, fan 106, exhaust system 108, or the like). In some implementations, the semiconductor processing tool 100 and/or the one or more components of the semiconductor processing tool 100 may include one or more devices 300 and/or one or more components of device 300. As shown in FIG. 3, device 300 may include a bus 310, a processor 320, a memory 330, a storage component 340, an input component 350, an output component 360, and a communication component 370.

[0030] Bus 310 includes a component that enables wired and/or wireless communication among the components of device 300. Processor 320 includes a central processing unit, a graphics processing unit, a microprocessor, a controller, a microcontroller, a digital signal processor, a field-programmable gate array, an application-specific integrated circuit, and/or another type of processing component. Processor 320 is implemented in hardware, firmware, or a combination of hardware and software. In some implementations, processor 320 includes one or more processors capable of being programmed to perform a function. Memory 330 includes a random access memory, a read only memory, and/or another type of memory (e.g., a flash memory, a magnetic memory, and/or an optical memory).

[0031] Storage component 340 stores information and/or software related to the operation of

[0032] device 300. For example, storage component 340 may include a hard disk drive, a magnetic disk drive, an optical disk drive, a solid state disk drive, a compact disc, a digital versatile disc, and/or another type of non-transitory computer-readable medium. Input component 350 enables device 300 to receive input, such as user input and/or sensed inputs. For example, input component 350 may include a touch screen, a keyboard, a keypad, a mouse, a button, a microphone, a switch, a sensor, a global positioning system component, an accelerometer, a gyroscope, and/or an actuator. Output component 360 enables device 300 to provide output, such as via a display, a speaker, and/or one or more light-emitting diodes. Communication component 370 enables device 300 to communicate with other devices, such as via a wired connection and/or a wireless connection. For example, communication component 370 may include a receiver, a transmitter, a transceiver, a modem, a network interface card, and/or an antenna.

[0033] Device 300 may perform one or more processes described herein. For example, a non-transitory computer-readable medium (e.g., memory 330 and/or storage component 340) may store a set of instructions (e.g., one or more instructions, code, software code, and/or program code) for execution by processor 320. Processor 320 may execute the set of instructions to perform one or more processes described herein. In some implementations, execution of the set of instructions, by one or more processors 320, causes the one or more processors 320 and/or the device 300 to perform one or more processes described herein. In some implementations, hardwired circuitry may be used instead of or in combination with the instructions to perform one or more processes described herein. Thus, implementations

described herein are not limited to any specific combination of hardware circuitry and software.

[0034] The number and arrangement of components shown in FIG. 3 are provided as an example. Device 300 may include additional components, fewer components, different components, or differently arranged components than those shown in FIG. 3. Additionally, or alternatively, a set of components (e.g., one or more components) of device 300 may perform one or more functions described as being performed by another set of components of device 300.

[0035] In this way, an exhaust system for a semiconductor processing tool may include one or more exhaust pipe segments arranged substantially around a processing tank, where the one or more or more exhaust pipe segments have openings to receive exhaust air to be provided to at least one exhaust output. As a result, airflow within the environment of the semiconductor processing tool may be uniformly distributed (e.g., such that the air flows in all directions around wafers being dried in the semiconductor processing tool, rather than a single direction), which causes a concentration of acid gas and/or a moisture concentration to be even among the wafers, thereby decreasing a defect rate in wafers processed by the semiconductor processing tool.

[0036] As described in greater detail above, some implementations described herein provide a semiconductor processing tool and an exhaust system of a semiconductor processing tool.

[0037] As described in greater detail above, some implementations described herein provide a semiconductor processing tool. The semiconductor processing tool includes a processing tank. The semiconductor processing tool includes an arm arranged to hold a plurality of wafers over the processing tank such that wafers in the plurality of wafers are horizontally stacked over the processing tank. The semiconductor processing tool includes a fan arranged over the arm to permit an airflow to be provided across surfaces of the wafers in a vertical direction toward the processing tank. The semiconductor processing tool includes an exhaust system including at least one exhaust output, and one or more exhaust pipe segments arranged substantially around the processing tank and connected to the at least one exhaust output, where the one or more exhaust pipe segments include a plurality of openings to receive exhaust air to be provided to the at least one exhaust output.

[0038] As described in greater detail above, some implementations described herein provide an exhaust system of a semiconductor processing tool. The exhaust system includes at least one exhaust output. The exhaust system includes one or more exhaust pipe segments connected to the at least one exhaust output, where the one or more exhaust pipe segments are substantially around a perimeter of a processing tank of the semiconductor processing tool, and where the one or more exhaust pipe segments include a plurality of openings, the plurality of openings being on one or more surfaces of the one or more exhaust pipe segments and being substantially around the perimeter of the processing tank.

[0039] As described in greater detail above, some implementations described herein provide a semiconductor processing tool. The semiconductor processing tool includes a processing tank. The semiconductor processing tool includes a fan arranged over the processing tank to cause an airflow to be provided toward the processing tank. The semiconductor processing tool includes an exhaust system

including one or more exhaust pipe segments arranged on at least two sides of the processing tank, where the one or more exhaust pipe segments include a plurality of openings to receive exhaust air to be provided to one or more exhaust outputs.

[0040] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. A semiconductor processing tool, comprising:
a processing tank;
one or more exhaust pipe segments, adjacent to the processing tank, comprising a plurality of openings on a bottom end of the one or more exhaust pipe segments to receive exhaust air; and
a fan arranged over the processing tank to permit an airflow to be provided across one or more wafers when the one or more wafers are between the fan and the processing tank.
2. The semiconductor processing tool of claim 1, further comprising:
at least one exhaust output to receive exhaust air from the one or more exhaust pipe segments and direct the exhaust air away from the semiconductor processing tool.
3. The semiconductor processing tool of claim 2, wherein the at least one exhaust output is a single exhaust output.
4. The semiconductor processing tool of claim 3, wherein the one or more exhaust pipe segments comprises a plurality of exhaust pipe segments that are each connected to the single exhaust output.
5. The semiconductor processing tool of claim 2, wherein the at least one exhaust output comprises a plurality of exhaust outputs.
6. The semiconductor processing tool of claim 5, wherein the one or more exhaust pipe segments comprises a plurality of exhaust pipe segments that are each connected to a different exhaust output of the plurality of exhaust outputs.
7. The semiconductor processing tool of claim 1, further comprising:
an arm to control a position of the one or more wafers above the processing tank.
8. The semiconductor processing tool of claim 7, wherein the arm is configured to submerge the one or more wafers into a solution in the processing tank, remove the one or more wafers from the processing tank, and hold the one or more wafers above the processing tank.

9. An exhaust system of a semiconductor processing tool, the exhaust system comprising:

at least one exhaust output; and

one or more exhaust pipe segments, connected to the at least one exhaust output, having a bottom surface comprising one or more openings.

10. The exhaust system of claim 9, wherein the one or more exhaust pipe segments comprises a single exhaust pipe segment around a processing tank.

11. The exhaust system of claim 9, wherein the one or more exhaust pipe segments comprises a plurality of exhaust pipe segments arranged around a processing tank.

12. The exhaust system of claim 11, wherein a gap is present along a side of the processing tank between a set of the plurality of exhaust pipe segments.

13. The exhaust system of claim 12, wherein a length of the gap is less than or equal to a length of a corresponding side of the processing tank.

14. The exhaust system of claim 11, wherein a gap is present at a corner of the processing tank between a first set of the plurality of exhaust pipe segments.

15. The exhaust system of claim 14, wherein a gap is present along a side of the processing tank between a second set of the plurality of exhaust pipe segments.

16. The exhaust system of claim 14, wherein a length of an exhaust pipe segment, of the set of the plurality of exhaust pipe segments, along a side of the processing tank is less than or equal to a length of a corresponding side of the processing tank.

17. A semiconductor processing tool, comprising:

a processing tank;

one or more exhaust pipe segments, arranged around the processing tank, comprising one or more openings on a bottom end of the one or more exhaust pipe segments to receive exhaust air,

wherein the processing tank extends beyond the bottom end of the one or more exhaust pipe segments;

a single pipe output connected to one of the one or more exhaust pipe segments; and

a fan extending above a top end of the one or more exhaust pipe segments.

18. The semiconductor processing tool of claim 17, wherein the one or more openings include a plurality of openings, and wherein a pattern of the plurality of openings is substantially symmetrical with respect to a center axis on a plane corresponding to the processing tank.

19. The semiconductor processing tool of claim 17, wherein the one or more openings include a plurality of openings evenly distributed around an entirety of a perimeter of the processing tank.

20. The semiconductor processing tool of claim 17, wherein the one or more exhaust pipe segments comprises a plurality of exhaust pipe segments arranged around a processing tank, wherein at least one of:

a gap is present at a corner of the processing tank between a first set of the plurality of exhaust pipe segments, or

a gap is present along a side of the processing tank between a second set of the plurality of exhaust pipe segments.

* * * * *